

# Christopher Musco

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## Academic Positions

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### Princeton University

*Research Instructor, Department of Computer Science*

**Princeton, NJ**

*2018 – present*

### Massachusetts Institute of Technology

*Research and Teaching Assistant, Theory of Computation Group*

**Cambridge, MA**

*2013 – 2018*

### Yahoo Labs

*Summer Research Intern, Scalable Machine Learning Group*

**New York, NY**

*2015*

## Research Areas

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foundations of data science • scalable machine learning • numerical linear algebra • theory of algorithms  
randomization • dimensionality reduction • sketching & streaming • spectral graph algorithms • optimization

## Education

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### Massachusetts Institute of Technology

*Ph.D. in Computer Science*

Advisor: Jonathan Kelner

Ph.D. Thesis: Faster Linear Algebra for Data Analysis and Machine Learning

M.S. Thesis: Dimensionality Reduction for Sparse and Structured Matrices

**Cambridge, MA**

*2013 – 2018*

### Yale University

*B.S. in Applied Mathematics, B.S. in Computer Science*

**New Haven, CT**

*2008 – 2012*

## Publications

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*Authors appear in alphabetical order, in the tradition of mathematics and theoretical computer science.*

**Sample Efficient Toeplitz Covariance Estimation.** Yonina Eldar, Jerry Li, Cameron Musco, and Christopher Musco. *In submission.* 2019.

**Low-Rank Approximation from Communication Complexity.** Cameron Musco, Christopher Musco, David Woodruff. *In submission.* 2019.

**Faster Spectral Sparsification in Dynamic Streams.** Michael Kapralov, Aida Mousavifar, Cameron Musco, Christopher Musco, Navid Nouri. *In submission.* 2019.

**A Universal Sampling Method for Reconstructing Signals with Simple Fourier Transforms.** Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *ACM Symposium on Theory of Computing (STOC)* 2019.

**Learning Networks from Random Walk-Based Node Similarities.** Jeremy Hoskins, Cameron Musco, Christopher Musco, Charalampos Tsourakakis. *Conference on Neural Information Processing Systems (NeurIPS)* 2018.

**Eigenvector Computation and Community Detection in Asynchronous Gossip Models.** Frederik Mallmann-Trenn, Cameron Musco, Christopher Musco. *International Colloquium on Automata, Languages, and*

*Programming (ICALP)* 2018.

**Minimizing Polarization and Disagreement in Social Networks.** Cameron Musco, Christopher Musco and Charalampos Tsourakakis. *The Web Conference (WWW)* 2018.

**Stability of the Lanczos Method for Matrix Function Approximation.** Cameron Musco, Christopher Musco, Aaron Sidford. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2018.

**Recursive Sampling for the Nyström Method.** Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2017.

**Random Fourier Features for Kernel Ridge Regression: Approximation Bounds and Statistical Guarantees.** Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *International Conference on Machine Learning (ICML)* 2017.

**Input Sparsity Time Low-Rank Approximation via Ridge Leverage Score Sampling.** Michael B. Cohen, Cameron Musco, Christopher Musco. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2017.

**Determining Tournament Payout Structures for Daily Fantasy Sports.** Christopher Musco, Maxim Sviridenko, Justin Thaler. *SIAM Algorithm Engineering and Experiments (ALENEX)* 2017.

**Principal Component Projection Without Principal Component Analysis.** Roy Frostig, Cameron Musco, Christopher Musco, Aaron Sidford. *International Conference on Machine Learning (ICML)* 2016.

**Randomized Block Krylov Methods for Stronger and Faster Approximate Singular Value Decomposition.** Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2015. Invited for full oral presentation (1 of 15 out of 403 accepted papers). Also presented at the 2016 Copper Mountain Conference on Iterative Methods

**Dimensionality Reduction for K-Means Clustering and Low Rank Approximation.** Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *ACM Symposium on Theory of Computing (STOC)* 2015.

**Principled Sampling for Anomaly Detection.** Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiroglou-Douskos, Martin Rinard. *Network and Distributed System Security Symposium (NDSS)* 2015.

**Uniform Sampling for Matrix Approximation.** Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *Innovations in Theoretical Computer Science (ITCS)* 2015.

**Single Pass Spectral Sparsification in Dynamic Streams.** Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *IEEE Symposium on Foundations of Computer Science (FOCS)* 2014, Special Issue SIAM Journal on Computing 2017.

## Teaching

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### **Advanced Algorithm Design (Princeton COS 521)**

*Instructor*

*Fall 2018*

### **Advanced Algorithms (MIT 6.854/18.415)**

*Teaching Assistant*

*Spring 2016*

### **Technical Communication Skills for Graduate Students (MIT 6.S977)**

*Workshop Leader*

*Spring 2016*

### **Introduction to Computer Science (Yale CPSC-201)**

*Teaching Assistant*

*Spring 2012*

### **Introduction to Programming (Yale CPSC-112)**

*Teaching Assistant*

*Fall 2011*

## Mentorship and Outreach

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### **SIAM/MathWorks Math Modeling Challenge**

*Judging and support*

*2018 – present*

Serve as a judge for SIAM's annual math modeling competition, which gives high school students an opportunity to apply mathematics, statistics, and data science to a real-world problem. Score student reports, provide feedback, and help with various tasks like problem writing.

### **MIT Graduate Communication Lab**

*Communication Adviser*

*2015 – 2018*

Helped to establish an EECS Communication Lab to provide over 700 graduate students with support in academic writing, oral presentation, job applications, and generally, in communicating effectively. Offered feedback, guidance, and coaching to individual students during weekly office hours.

### **CSAIL Algorithms Office Hours**

*Member*

*2016 – 2018*

Advised researchers, generally in applied fields, on framing and solving algorithmic problems arising in their work.

### **Research Science Institute at MIT**

*Summer Research Adviser*

*2014*

Primary supervisor for a high school student participating in the prestigious RSI summer research program. Advisee's project, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

## Talks and Presentations

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### **Learning Signals with Simple Fourier transforms**

Columbia University, Theory Seminar

*Oct. 2018*

UC Berkeley, Simons Institute Workshop

*Sept. 2018*

Princeton University, Theory Seminar

*Sept. 2018*

### **The Lanczos Method in Data Science: Challenges and Opportunities**

Georgetown University, CS Seminar

*Feb. 2018*

Symposium on Discrete Algorithms (SODA)

*Jan. 2018*

### **Unifying Random Fourier Features and Leverage Scores for Kernel Matrix Approximation**

Institute for Advanced Study, Computer Science and Discrete Mathematics Seminar

*Feb. 2018*

Harvard University, guest lecture in "Sketching Algorithms for Big Data"

*Nov. 2017*

Microsoft Research, Redmond

*Sept. 2017*

### **Sketching and Sampling Methods for Large Scale Linear Algebra**

Twitter, Cambridge

*July 2017*

### **Recursive Sampling for the Nyström Method**

Neural Information Processing Systems (NeurIPS) poster session

*Dec. 2017*

New England Machine Learning Day poster session

*May 2017*

### **Algorithms for Determining Tournament Payout Structures**

Meeting on Algorithm Engineering and Experiments (ALENEX)

*Jan. 2017*

### **Introduction to Linear Sketching**

MIT, Theory Retreat

*Sept. 2016*

### **Principal Component Projection without Principal Component Analysis**

International Conference on Machine Learning (ICML)

*June 2016*

### **Ridge Leverage Score Sampling**

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|----------------------------------------------------------------|-----------|
| National Institute of Informatics, Shonan Meeting              | July 2016 |
| MIT, Algorithms & Complexity Seminar                           | July 2016 |
| University of Utah, Data Group Meeting                         | Jan. 2016 |
| <b>Randomized Block Krylov Methods</b>                         |           |
| Neural Information Processing Systems (NeurIPS) poster session | Dec. 2015 |
| <b>Large Scale Column Subset Selection</b>                     |           |
| Yahoo Labs, Science Week poster session                        | July 2015 |
| <b>Dimensionality Reduction for k-Means Clustering</b>         |           |
| IBM T.J. Watson, Mathematical Sciences research seminar        | Aug. 2015 |
| Symposium on Theory of Computing (STOC)                        | June 2015 |
| MIT, Theory Lunch                                              | June 2014 |
| <b>Principled Sampling for Anomaly Detection</b>               |           |
| MIT, Defense Advanced Research Projects Agency site visit      | Mar. 2015 |
| Network and Distributed System Security Symposium (NDSS)       | Feb. 2015 |
| <b>Uniform Sampling for Matrix Approximation</b>               |           |
| Yahoo Labs, Scalable Machine Learning research seminar         | June 2015 |
| <b>Single Pass Spectral Sparsification in Dynamic Streams</b>  |           |
| MIT, Annual Sublinear Algorithms Day poster session            | Apr. 2015 |
| Harvard University, Theory Seminar                             | Nov. 2014 |
| Foundations of Computer Science (FOCS)                         | Oct. 2014 |

## Service

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| Conference Programming.....                                                                                                                                                                                                                                                                                                                                                                          |  |
| NeurIPS 2019 Area Chair                                                                                                                                                                                                                                                                                                                                                                              |  |
| Conference Reviewer.....                                                                                                                                                                                                                                                                                                                                                                             |  |
| COLT (2016, 2018, 2019), ESA (2015, 2017), FOCS (2015,2018), ICALP (2015, 2017, 2018), ICML (2018, 2019) ITCS (2018), IWOCA (2018), NeurIPS (2015, 2016, 2017, 2018), NDSS (2015), RANDOM (2017, 2018), STACS (2018), SODA (2016, 2017, 2018, 2019), STOC (2018, 2019), WALCOM (2018)                                                                                                                |  |
| Journal Reviewer.....                                                                                                                                                                                                                                                                                                                                                                                |  |
| SIAM Journal on Matrix Analysis and Applications (2016, 2017), SIAM Journal on Scientific Computing (2019), Science Advances (2018), Mathematical Programming (2018), IEEE Transactions on Signal Processing (2016, 2017, 2018), Journal of Machine Learning Research (2018, 2019), Advances in Data Analysis and Classification (2018), IEEE Transactions on Knowledge and Data Engineering (2017), |  |
| Notables.....                                                                                                                                                                                                                                                                                                                                                                                        |  |
| ICML 2018 Outstanding Reviewer                                                                                                                                                                                                                                                                                                                                                                       |  |

## Other Research Experience

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|-------------------------------------------------------------------|------|
| <b>UC Berkeley, Simons Institute for the Theory of Computing</b>  |      |
| <i>Invited participant</i>                                        | 2018 |
| Workshop on Randomized Numerical Linear Algebra and Applications. |      |
| <b>Japan National Institute of Informatics</b>                    |      |
| <i>Invited participant</i>                                        | 2016 |

Shonan meeting on Recent Advances in Randomized Numerical Linear Algebra.

## Honors and Awards

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**National Science Foundation** 2014-2017

*Graduate Research Fellowship, recipient.*

**Yale University** 2012

*Cum Laude, Distinction in Both Majors.*

## Grants and Funding

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Travel Awards.....

SIAM Early Career 2019, NeurIPS 2018, NeurIPS 2015

## Other Professional Experience

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**Redfin** **Seattle, WA**

*Software Engineer, Data Team* 2012 – 2014

Developed backend Java infrastructure for internet-powered real estate startup. Incorporated large-scale machine learning algorithms into production code. Managed three interns and onboarded four new-hires.

**Elysium Digital** **Cambridge, MA**

*Summer Consultant* 2011

Provided technical expertise in software patent cases at leading litigation consulting company.

**Amicus** **New Haven, CT**

*Software Developer* 2010 – 2011

Built applications for political campaign management and social fundraising at an early-stage startup that went on to raise 4 million dollars in seed funding.